



High Performance Vision-Guided Rocket Tracker

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TEAM #3 SPACEY | MCMASTER UNIVERSITY SFWRENG 4G06 | MCMASTER ROCKETRY TEAM



THE CHALLENGE

Rockets reach **Mach 3** in seconds.

Manual optical tracking is physically impossible for small-profile hobbyist rocketry. Commercial rigs cost upwards of \$10,000.

100 km

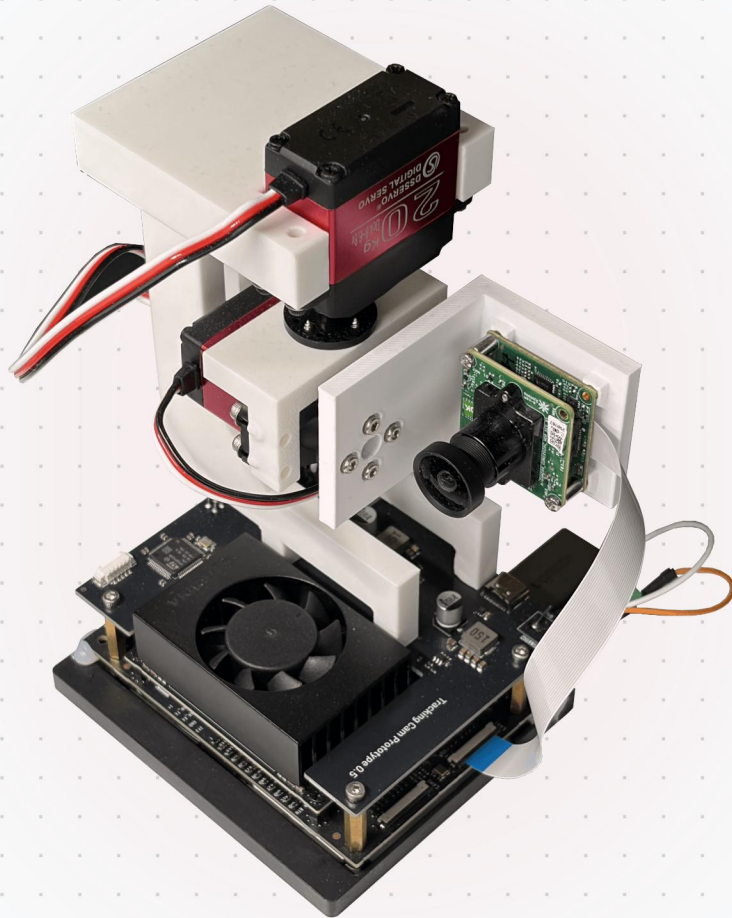
MAX ALTITUDE

>10 g

ACCELERATION

PRESENTATION ROADMAP

- 1. Problem & Solution
- 2. Architecture & Live Demo
- 3. Data, CV & ML Pipeline
- 4. Hardware & UI Control
- 5. Testing, Challenges & Future



Autonomous. Accessible. Accurate.

A fully autonomous, closed-loop optical tracking system engineered to meet strict performance constraints.



Autonomous Edge AI Tracking

Works in remote, offline, and hazardous conditions



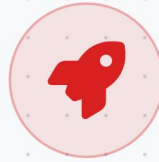
Accurate and Precise

60Hz closed-loop control and digital stabilization



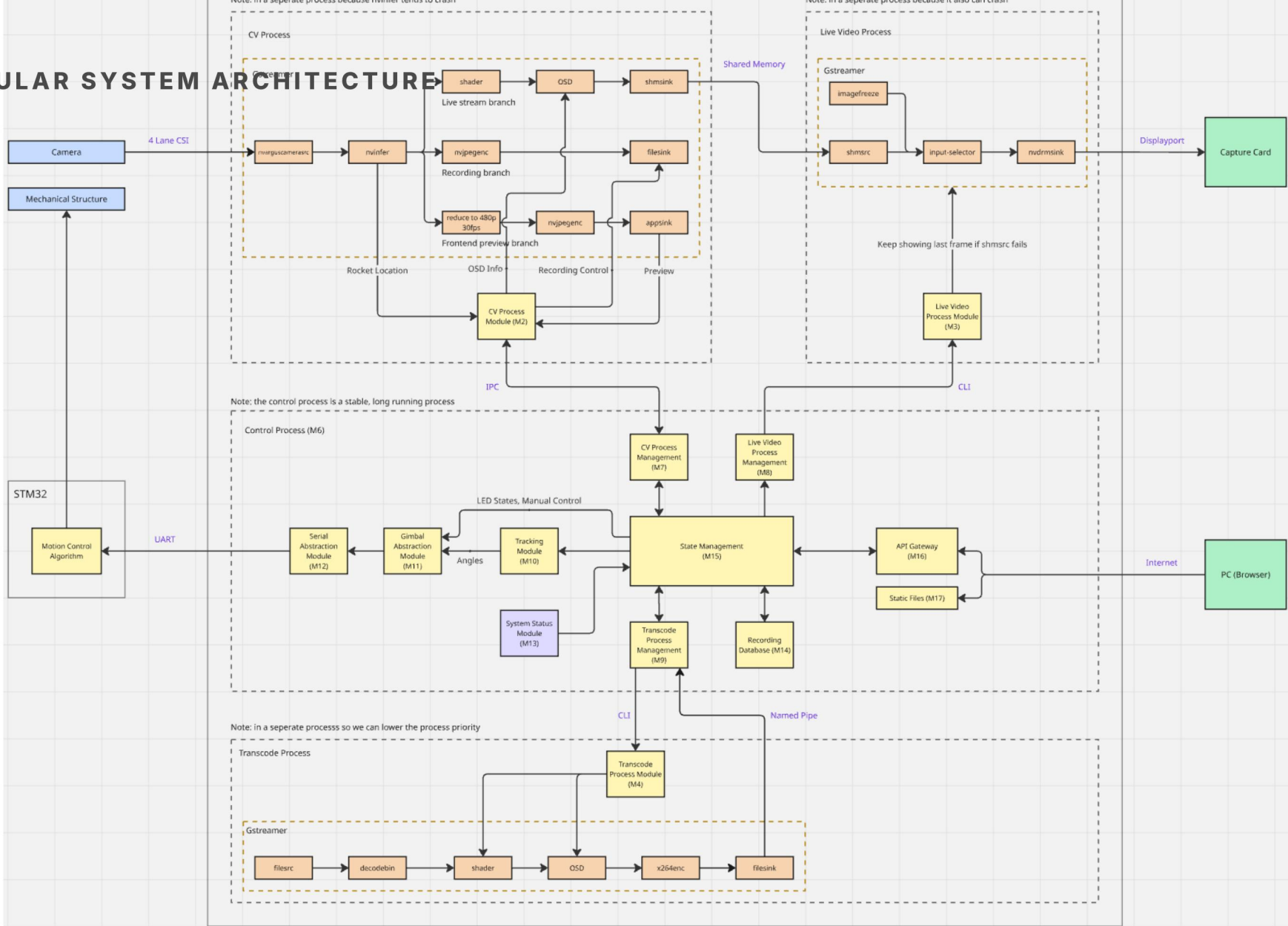
Low Cost

Total BOM under \$500 CAD (vs \$10k+ commercial)



LIVE DEMO

MODULAR SYSTEM ARCHITECTURE



Custom labelling platform for rocket data.

All training data flows through our in-house **RoCam Labeler** — a full-stack annotation tool built from scratch to handle our unique dataset requirements.



SAM-Assisted Annotation

One-click segmentation masks via Meta SAM



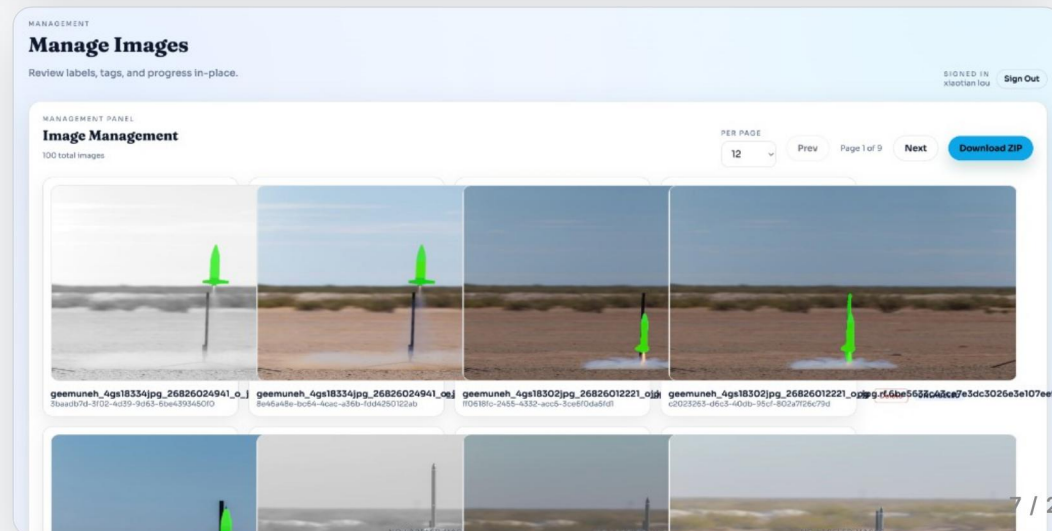
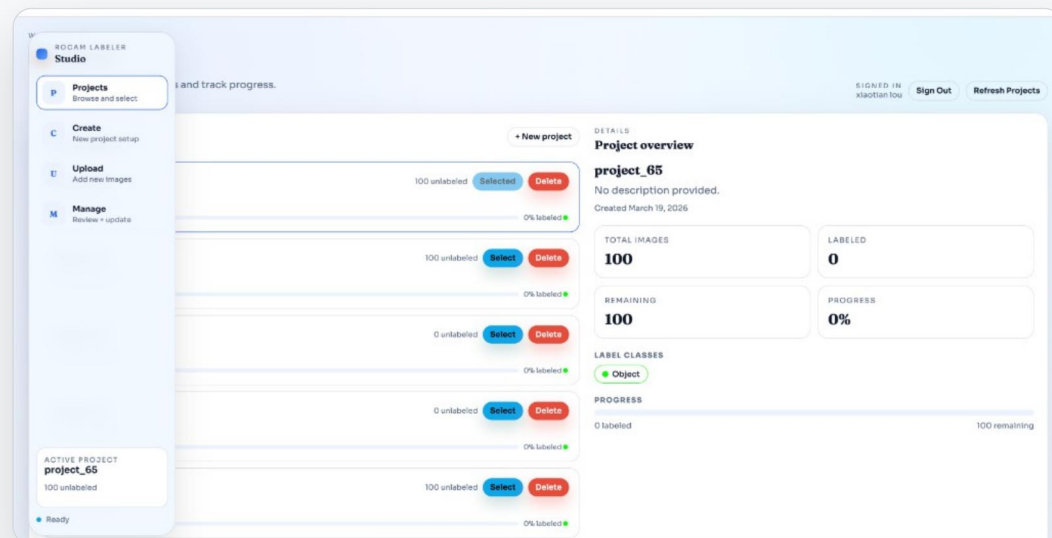
Project Management

Multi-project workspace with progress tracking

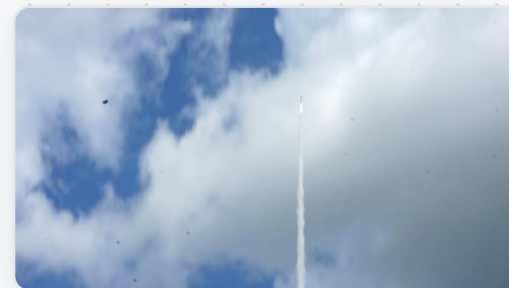


Batch Upload / Export

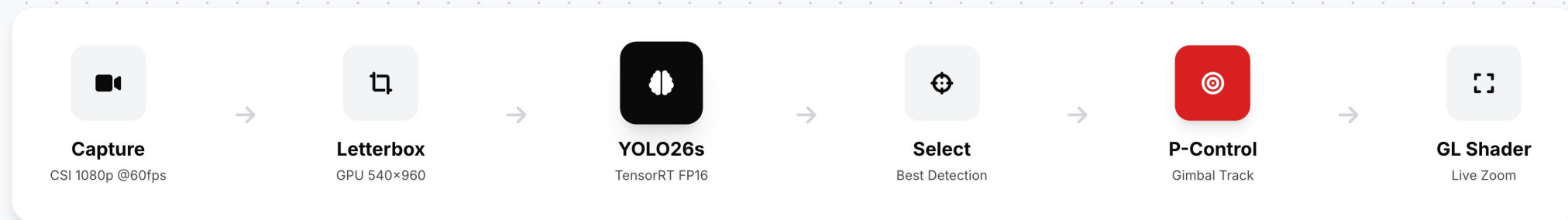
ZIP-based bulk import and YOLO-format export



Optimized for extreme distance and strict latency budgets.



Raw frame — can you spot the rocket?



🔍 Why YOLO26s?

One-to-one detection head eliminates NMS entirely. 9.47M parameters engineered for FP16—only 9% accuracy loss on Jetson vs. 30% for alternative architectures.

🔥 Full GPU Pipeline

DeepStream runs decode, letterbox, and TensorRT inference entirely on GPU. Confidence threshold filters noise; probe selects single best detection per frame.

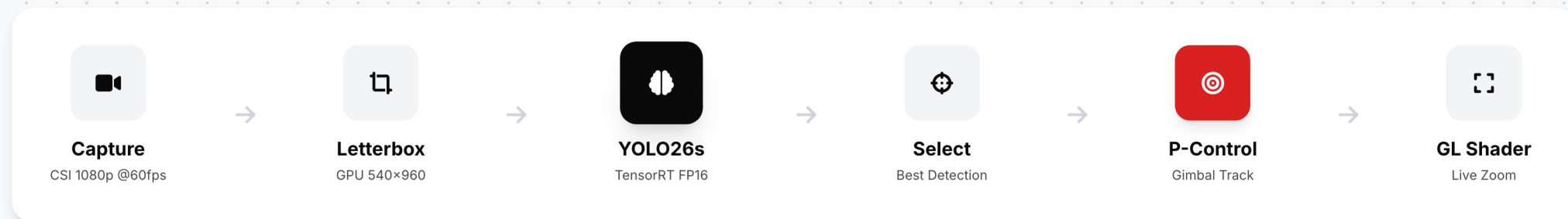
🎯 Instant Visual Lock

GL fragment shader crops and zooms on target within one display frame. P-control drives gimbal via UART. Zero perceived tracking latency.

Optimized for extreme distance and strict latency budgets.



Target <0.1% of frame — detected at conf 0.52



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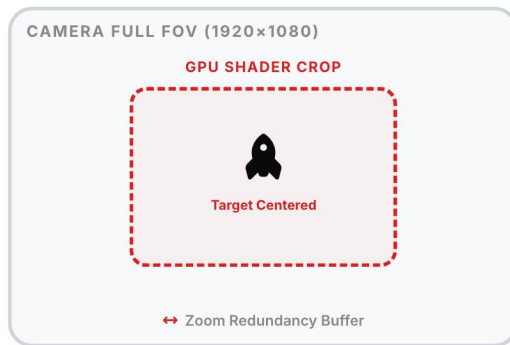
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Optical redundancy enables perceptually instant tracking.



FOV Redundancy Model

Wide optical view guarantees target stays in frame while gimbal mechanically catches up



Detection Result

BBox (cx, cy, w, h) from YOLO @ 60fps

~1MS



GPU Shader Path

- 1 Compute tx, ty, scale from bbox
- 2 GL shader crops & centers frame
- 3 Livestream output → operator sees instant lock

~100MS



Gimbal Servo Path

- 1 P-control → delta pan/tilt
- 2 UART → STM32 → PWM servo
- 3 Mechanical motion re-centers optical FOV



Zero Perceived Latency

Operator sees target locked within one display frame (~16ms). Gimbal catches up transparently.

96.2%

PEAK TRAINING ACCURACY

mAP@50 across 10+ experiments on H100

9% Deployment Gap

FP16 quantization on Jetson Orin Nano. Our best-in-training architecture lost **30%** at deployment—the production recipe held.

94.3%
PRECISION

89.8%
RECALL

60
FPS PIPELINE

9.47M
PARAMETERS

20.5
GFLOPS

<1ms
SHADER LATENCY

Aggressive Augmentation Training

YOLO26s trained end-to-end for **420 epochs** on H100 GPU. Data labelled via our custom **Labelling App** with SAM-assisted annotation.



Full Mosaic = 1.0 (Key Finding)

Every image trained as 2×2 mosaic, quadrupling effective object density. Disabled last 80 epochs for full-resolution refinement. All reduced-mosaic variants failed.



15,832 Images + 4,000 COCO Hard Negatives

25% of training set is pure background from COCO, forcing a harder decision boundary to suppress false positives in real-world deployment.



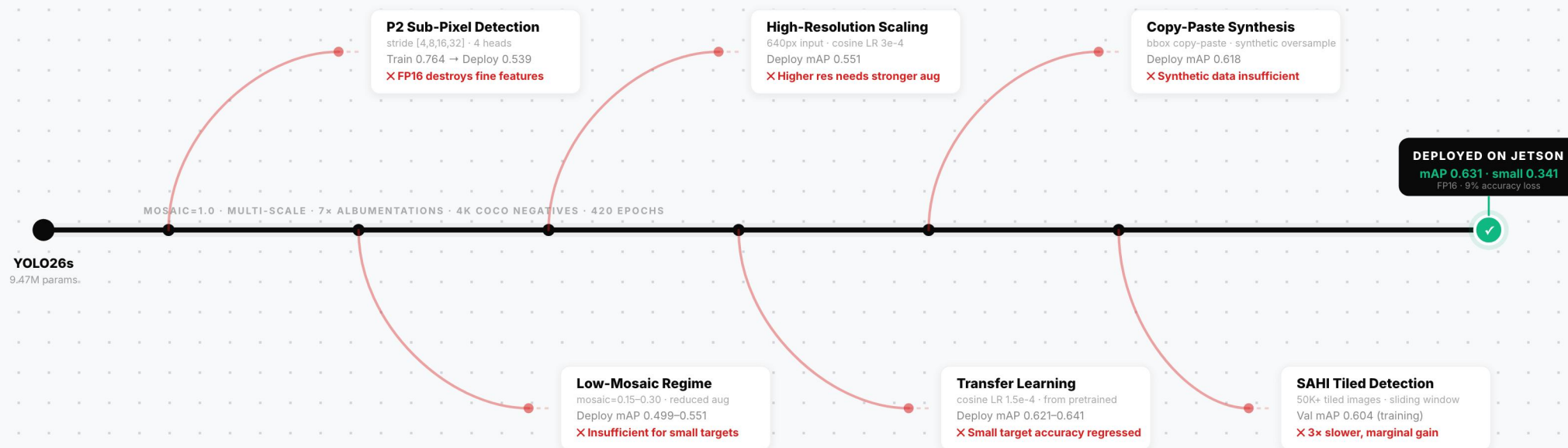
Multi-Scale + 7 Albumentations

Variable resolution (544–960px), 180° rotation, motion blur, compression artefacts, and sensor noise transforms for real-world robustness.

MODEL DEVELOPMENT JOURNEY

Systematic Exploration. **One Survivor.**

Every promising architecture and training strategy tested against real edge deployment constraints



10+ EXPERIMENTS
CONDUCTED

100+ GPU-HOURS
ON H100

9% FP16 ACCURACY
LOSS (BEST)

1 SURVIVED
DEPLOYMENT

Bare-metal **Rust** for hard real-time execution.

60 Hz

CONTROL LOOP

Embassy

RUST ASYNC

CRC-8

DATA INTEGRITY

<0.5s

WATCHDOG TRIP



Jetson Orin

Vision / Math

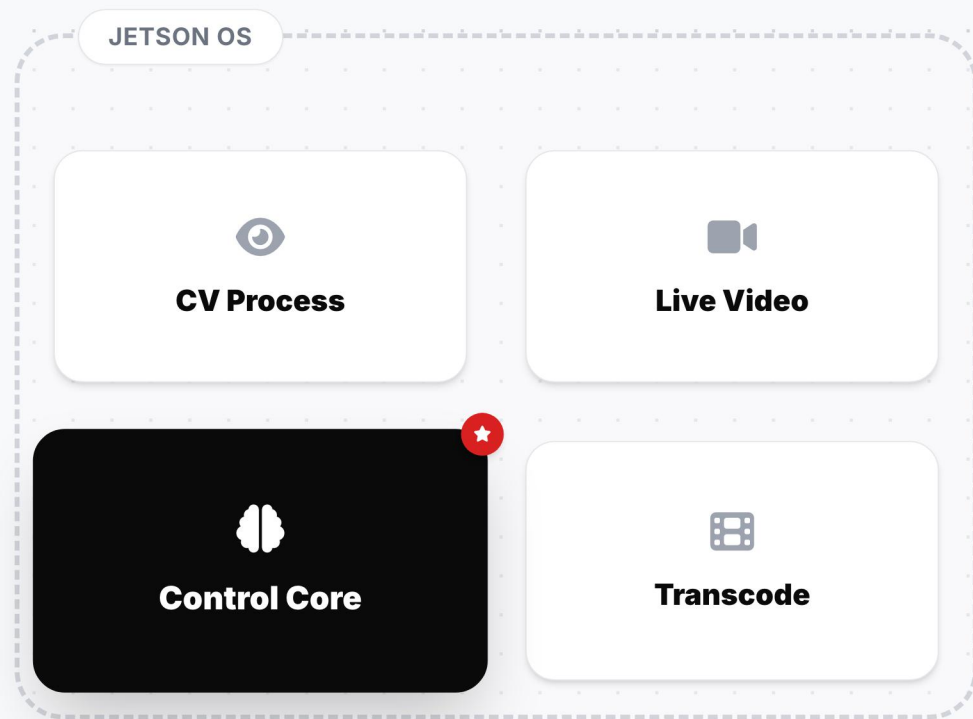
115200 BAUD UART

STM32F405

Custom PCB



— FAULT-TOLERANT ARCHITECTURE



Built for isolation and real-time survival.



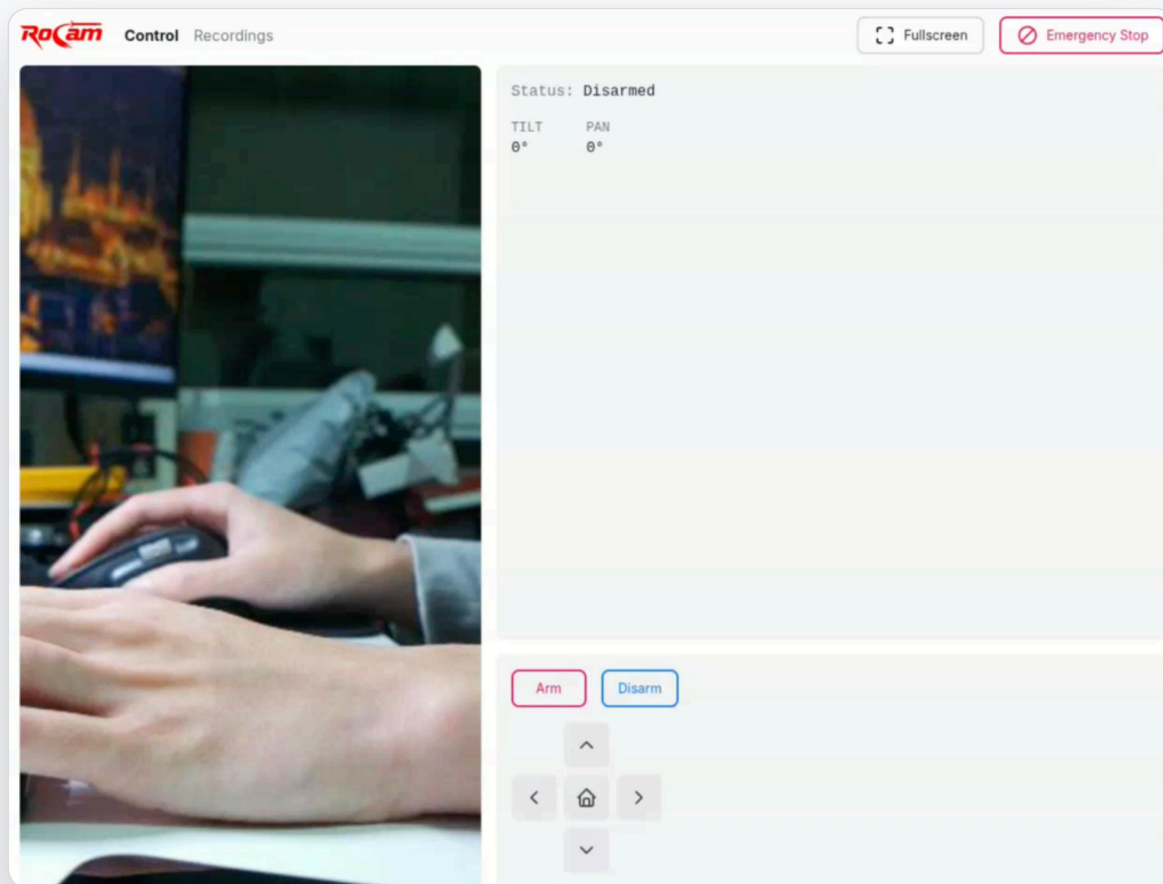
Independent Restart

If the CV inference crashes, the control process maintains connection to the gimbal, preventing lock-up.



Zero-copy IPC

Shared-memory architecture allows massive 1080p frames to move between processes instantly.



Command center in a browser.

REACT • TYPESCRIPT • VITE



Low-latency live preview



Direct manual gimbal override



Real-time health telemetry



Full Bilingual (EN/FR) support

— RECORDING WORKFLOW

FROM LAUNCH TO REPLAY

Record raw. Rebuild smart.

RoCam stores the raw camera feed and per-frame telemetry separately, then reconstructs stabilized playback and downloads on demand.

TRIGGER

Browser UI

Start / stop from
Mission Control

STORAGE

AVI + JSONL

Video frames plus OSD
telemetry

OUTPUT

Preview / Download

Rendered after the flight
event

WHY THIS SPLIT MATTERS

-  Capture stays lightweight during launch instead of doing heavy replay rendering in the critical path.
-  The saved telemetry lets us re-apply OSD, timestamps, zoom transforms, and stabilization later for post-flight review.

1. CAPTURE



Raw video branch to **video.avi**

The CV pipeline swaps in a recording sink and writes the camera stream without waiting for replay rendering.



2. LOG



Per-frame OSD data to **log.txt**

Each frame stores pan, tilt, focal length, FPS, timestamps, and stabilization transforms as JSONL telemetry.



3. REPLAY



Separate transcode for stabilized review

Preview and download are reconstructed later with the same OSD and shader path, producing a polished replay stream for analysis.

481

AUTOMATED TESTS PASSED



Backend (pytest)

313 Unit Tests

88% Coverage



Frontend (Vitest)

168 Component Tests

86% Coverage



System E2E

GitHub Actions CI

100% Pass Rate

— PROJECT CHALLENGES

Real-time Latency

Python's GIL blocking CV loops.



Solution

Shared-memory IPC bypassing GIL entirely.

Micro-Object Detection

Rockets vanishing into background noise.



Solution

YOLO26s-P2 Head & Hard-negative mining.

Hardware Sync

UART dropping packets at 115200 baud.



Solution

Logic analyzer debug + custom CRC-8 checksums.

Designed to scale from 200m hobby rockets to 3km+ high-power launches.



Performance

- 4K @ 60fps tracking
- TensorRT INT8 Quantization
- Optical Auto-focus



Hardware Scaling

- Full-size Gimbal Mounts
- Multi-camera Arrays
- Weatherproof Enclosures

NEXT STEP



Software UX

- Mobile-native Apps
- Post-flight Analytics
- AR Telemetry Overlay

— STAKEHOLDER FEEDBACK

MCMaster Rocketry Team Review

Built with **operator feedback**, not just lab assumptions.

STAKEHOLDER REQUEST

"A more direct and intuitive way to move the gimbal."

Captured in the stakeholder walkthrough and turned into drag-to-move control on the live preview.



Direct Control

Drag-to-move framing added beside the D-pad.



Clear Status

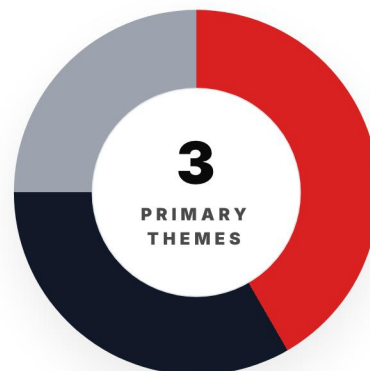
More visible armed, target-lost, storage, and network feedback.



Stable Recording

Pause/resume behavior refined for download and recording actions.

FEEDBACK THEMES



- **More direct camera framing**
Stakeholders pushed for faster, more intuitive setup before launch.
- **Clear operator state visibility**
Armed, tracking, storage, device, and network states needed to be obvious at a glance.
- **Reliable recording management**
Playback and download flows were tightened so recording interactions feel safer and less fragile.

EVIDENCE IN V&V

UI

Walkthrough-driven changes

OPS

Better situational awareness

REC

More robust recording workflow



Thank You.

Questions?



<https://github.com/SpaceY-Labs/RoCam>

WITH THANKS TO



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**Dr. Shahin
Sirouspour**

Faculty guidance and
project support



INSTRUCTOR

**Dr. Spencer
Smith**

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